

Introduction to Contract Theory and Auctions

Knowledge and Beliefs

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ECON 403/503
Fall 2022

Definition 1.1

A **knowledge space** is a tuple $(N, \Omega, \{T_i\}_{i \in N}, \{\tau_i\}_{i \in N})$ where

- 1 N is the set of players.
- 2 Ω is the (*finite*) set of **states of the world**.
- 3 T_i is the space of possible **types** of player i .
- 4 $\tau_i : \Omega \rightarrow T_i$ is player i 's **private signal** or **type**.

Definition 1.2

Player i 's **information partition function** $P_i : \Omega \rightarrow 2^\Omega$ is defined by

$$P_i(\omega) = \{\omega' : \tau_i(\omega') = \tau_i(\omega)\} = \tau_i^{-1}(\tau_i(\omega)).$$

The collection $\mathcal{P}_i = \{P_i(\omega)\}_{\omega \in \Omega}$ is called **i 's information partition**.

Remark

Knowledge spaces can be also given in terms of partitions, i.e., $N(, \Omega, \{\mathcal{P}_i\}_{i \in N})$.

Partitions

A partition of Ω is a family $\mathcal{P}$ of pairwise-disjoint non-empty subsets of Ω whose union includes Ω . That is, a partition $\mathcal{P}$ satisfies

P1 $\bigcup \mathcal{P} = \bigcup_{P \in \mathcal{P}} P = \Omega.$

P2 $A, B \in \mathcal{P}, A \cap B \neq \emptyset \Rightarrow A = B.$

Example 1.3

Ω is the set of all possible outcomes of tossing a pair of dice:

$$\Omega = \{(k, l) : k, l = 1, \dots, 6\}.$$

$N = \{1, 2\}$, Player 1 observes the first die, Player 2 observes the sum: $T_1 = \{1, 2, \dots, 6\}$, $\tau_1(\omega) = \tau_1(k, l) = k$; and $T_2 = \{2, \dots, 12\}$, $\tau_2(\omega) = \tau_2(k, l) = k + l$.

Example 1.4

Hats riddle, before the announcement. $\Omega = \{r, b\}^n$, $T_i = \{r, b\}^{n-1}$, and $\tau_i(\omega) = \omega_{-i}$.

Consider a knowledge space $(N, \Omega, \{T_i\}_{i \in N}, \{\tau_i\}_{i \in N})$. Suppose $A \in 2^\Omega$ is an event.

Definition 1.5

Player i **knows A at ω** if $P_i(\omega) \subseteq A$, or equivalently if $\tau_i(\omega') = \tau_i(\omega)$ implies $\omega' \in A$.

Definition 1.6

Define $K_i A$ as the set of states of the world in which i knows A :

$$\begin{aligned} K_i A &= \{\omega : P_i(\omega) \subseteq A\} \\ &= \bigcup \{P_i(\omega) : P_i(\omega) \subseteq A\}. \end{aligned}$$

Theorem 1.7 (Kripke's S5 system)

- 1 $K_i\Omega = \Omega$. A player knows that some state of the world has occurred.
- 2 $K_iA \cap K_iB = K_i(A \cap B)$. A player knows A and a player knows B if and only if she knows A and B .
- 3 Axiom of knowledge. $K_iA \subseteq A$. If a player knows A then A has indeed occurred.
- 4 Axiom of positive introspection. $K_iK_iA = K_iA$. If a player knows A , then she knows that she knows A .
- 5 Axiom of negative introspection. $(K_iA)^c = K_i((K_iA)^c)$. If a player does not know A then she knows that she does not know A .

Remark

Let $L : 2^\Omega \rightarrow 2^\Omega$ be map satisfying S5. Then, there exists a type space T and a function $\tau : \Omega \rightarrow T$ such that L is equal to the associated knowledge operator given by

$$KA = \{\omega : P(\omega) \subseteq A\}$$

where $P(\omega) = \tau^{-1}(\tau(\omega))$.

Definition 1.8

An event $A \subseteq 2^\Omega$ is said to be **self-evident** to player i if $K_i A = A$. Let Σ_i be the collection of self-evident events.

Example 1.9 (Color-blindness, extended)

Recall: $\Omega = \{\omega_{g1}, \omega_{g2}, \omega_{r1}, \omega_{r2}, \omega_{p1}, \omega_{p2}\}$, $\mathcal{P}_A = \{\{\omega_{g1}, \omega_{g2}, \omega_{r1}, \omega_{r2}\}, \{\omega_{p1}, \omega_{p2}\}\}$, $\mathcal{P}_B = \{\{\omega_{g1}, \omega_{r1}\}, \{\omega_{g2}\}, \{\omega_{r2}\}, \{\omega_{p1}\}, \{\omega_{p2}\}\}$. Then

$\Sigma_A = \{\emptyset, \{\omega_{p1}, \omega_{p2}\}, \{\omega_{g1}, \omega_{g2}, \omega_{r1}, \omega_{r2}\}, \Omega\}$,

$\Sigma_B = \{\emptyset, \{\omega_{g2}\}, \{\omega_{p1}\}, \{\omega_{p2}\}, \{\omega_{r2}\}, \{\omega_{g1}, \omega_{r1}\}, \{\omega_{g2}, \omega_{p1}\}, \{\omega_{g2}, \omega_{p2}\}, \{\omega_{g2}, \omega_{r2}\}, \{\omega_{p1}, \omega_{p2}\}, \{\omega_{p1}, \omega_{r2}\}, \{\omega_{p2}, \omega_{r2}\}, \{\omega_{g1}, \omega_{g2}, \omega_{r1}\}, \{\omega_{g1}, \omega_{p1}, \omega_{r1}\}, \{\omega_{g1}, \omega_{p2}, \omega_{r1}\}, \{\omega_{g1}, \omega_{r1}, \omega_{r2}\}, \{\omega_{g2}, \omega_{p1}, \omega_{p2}\}, \{\omega_{g2}, \omega_{p1}, \omega_{r2}\}, \{\omega_{g2}, \omega_{p2}, \omega_{r2}\}, \{\omega_{p1}, \omega_{p2}, \omega_{r2}\}, \{\omega_{g1}, \omega_{g2}, \omega_{p1}, \omega_{r1}\}, \{\omega_{g1}, \omega_{g2}, \omega_{p2}, \omega_{r1}\}, \{\omega_{g1}, \omega_{g2}, \omega_{r1}, \omega_{r2}\}, \{\omega_{g1}, \omega_{p1}, \omega_{p2}, \omega_{r1}\}, \{\omega_{g1}, \omega_{p1}, \omega_{r1}, \omega_{r2}\}, \{\omega_{g1}, \omega_{p2}, \omega_{r1}, \omega_{r2}\}, \{\omega_{g2}, \omega_{p1}, \omega_{p2}, \omega_{r2}\}, \{\omega_{g1}, \omega_{g2}, \omega_{p1}, \omega_{p2}, \omega_{r1}\}, \{\omega_{g1}, \omega_{g2}, \omega_{p1}, \omega_{r1}, \omega_{r2}\}, \{\omega_{g1}, \omega_{g2}, \omega_{p2}, \omega_{r1}, \omega_{r2}\}, \{\omega_{g1}, \omega_{p1}, \omega_{p2}, \omega_{r1}, \omega_{r2}\}, \Omega\}$

Definition 1.10

A **field of sets** is a pair $(\Omega, \mathcal{A})$, where **$\mathcal{A}$ is an algebra over Ω** , which is a family of subsets with the properties:

A1 $\emptyset \in \mathcal{A}$ and $\Omega \in \mathcal{A}$.

A2 If $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$.

A3 If $A, B \in \mathcal{A}$, then $A \cap B \in \mathcal{A}$ and $A \cup B \in \mathcal{A}$.

By induction, $\mathcal{A}$ is closed under finite intersections and finite unions.

A **σ -field** is a field equipped with a **σ -algebra**, that is, closed under countable intersections and countable unions. A **complete field** is a field equipped with an algebra closed under arbitrary intersections and unions.

Lemma 1.11

The intersection of a family of fields is a field.

The intersection of a family of σ -fields is a σ -field.

The intersection of a family of complete fields is a complete field.

Thus, given a family $\mathcal{B}$ of subsets, there is a smallest field $\mathcal{F}(\mathcal{B})$ that includes $\mathcal{B}$, a smallest σ -field $\sigma(\mathcal{B})$ that includes $\mathcal{B}$, and a smallest complete field $\mathcal{F}^(\mathcal{B})$ that includes $\mathcal{B}$.*

Definition 1.12

An **atom** of a field is an element of the field that cannot be written as the disjoint union of two nonempty elements of the field.

Proposition 1.13

Let $\mathcal{P}$ be a partition. For any A , define $P(A) = \bigcup \{P(\omega) : \omega \in A\}$. A set A belongs to $\mathcal{F}^(\mathcal{P})$ if and only if $P(A) = A$. The set A is an atom of $\mathcal{F}^*(\mathcal{P})$ if and only if $A \in \mathcal{P}$.*

Proposition 1.14

If $\mathcal{F}$ is a complete field of subsets of Ω , the atoms of $\mathcal{F}$ constitute a partition of Ω .

Recall that Σ_i is the collection of self-evident events for player i . Let $A \in 2^\Omega$.

Claim 1.15

- ① A is self-evident if and only if it is the union of partition elements $P_i(\omega)$
- ② Σ_i is an algebra over Ω , and the field (Ω, Σ_i) coincides with $\mathcal{F}^*(\mathcal{P}_i)$, the complete field generated by $\mathcal{P}_i$.

Remark

An alternative to specifying type sets or partitions, we can start by specifying the self-evident algebras Σ_i . We can recover $P_i(\omega)$ (or T_i and τ_i) from Σ_i by

$$P_i(\omega) = \bigcap \{S \in \Sigma_i : \omega \in S\}.$$

Thus, a knowledge space can be described by $(N, \Omega, \{\Sigma_i\}_{i \in N})$.

Proposition 1.16

Let $A \in 2^\Omega$. Then $K_i A \in \Sigma_i$, and in particular

$$K_i A = \bigcup \{S \in \Sigma_i : S \subseteq A\}.$$

Since any event algebra is closed to unions, it follows that $K_i A$ is the largest element of Σ_i that is contained in A :

$$K_i A = \max\{S \in \Sigma_i : S \subseteq A\}.$$

Definition 1.17

Let $(N, \{T_i\}_{i \in N}, \{\tau_i\}_{i \in N})$ be a finite knowledge space. An event A is said to be **common knowledge at $\omega \in \Omega$** if for any sequence $i_1, i_2, \dots, i_k \in N$ it holds that

$$\omega \in K_{i_1} K_{i_2} \cdots K_{i_k} A$$

Definition 1.18

Let Σ, Π be two sub-algebras of some algebra. We say that Σ is a **refinement** of Π if $\Pi \subseteq \Sigma$. In this case we also say that Π is a coarsening of Σ .

Definition 1.19

The **meet** of two algebras Σ_1 and $\Sigma_2 \subseteq \Sigma$ is the finest sub-algebra of Σ that is a coarsening of each Σ_i . Their **join** is the coarsest sub-algebra of Σ that is a refinement of each Σ_i .

An alternative description of Common Knowledge events

Let $\Sigma_C = \bigcap_{i \in N} \Sigma_i$.

Claim 1.20

The following are equivalent:

- ① $C \in \Sigma_C$.
- ② $K_i C = C$ for all $i \in N$.

Recall that

$$K_i A = \bigcup \{S \in \Sigma_i : S \subseteq A\} = \max\{S \in \Sigma_i : S \subseteq A\}.$$

Similarly we define:

$$K_C A = \bigcup \{S \in \Sigma_C : S \subseteq A\} = \max\{S \in \Sigma_C : S \subseteq A\}.$$

- It is useful to think about common knowledge in terms of a graph¹, which we will call G_C .
- The vertices of G_C are elements of Ω , and there is an edge between $\omega, \omega' \in \Omega$ if there exists a player i such that $P_i(\omega) = P_i(\omega')$.
- $C(\omega)$ denotes the **connected component of ω** , i.e., the maximal set of vertices in the graph for which there exists a path² from ω .

¹An undirected graph G , is a collection (V, E) where V is a set of vertices and E is a set of edges between those vertices.

²A path from ω to ω' is a sequence of edges $(\omega_1, \omega_2), (\omega_2, \omega_3), \dots, (\omega_{k-1}, \omega_k)$ such that $\omega_1 = \omega$ and $\omega_k = \omega'$.

Theorem 1.21

Given a finite knowledge space $(N, \Omega, \{\Sigma_i\}_{i \in N})$, a state of the world $\omega \in \Omega$ and an event A , the following are equivalent:

- ① *A is common knowledge at ω .*
- ② *$C(\omega) \subseteq A$.*
- ③ *$\omega \in K_C A$.*

The hats riddle

Recall $\Omega = \{r, b\}^n$, $T_i = \{r, b\}^{n-1}$, and $\tau_i(\omega) = \omega_{-i}$. The information partition functions are $P_i(\omega) = \{(\omega_{-i}, r), (\omega_{-i}, b)\}$. Thus the set of edges of G_C is

$$E = \{P_i(\omega)\}_{i,\omega} = \{(\omega_{-i}, r), (\omega_{-i}, r)\} : \omega_{-i} \in \{r, b\}^{n-1}\}$$

Claim 1.22

$$\Sigma_C = \{\emptyset, \Omega\}.$$

A **belief** of a player over a finite space Ω is a probability measure on Ω

Example 1.23

Consider a monopoly seller (S) that would like to sell a single object to a single buyer (B). The value of the product to the buyer is $v \geq 0$. The seller sets a price $p \geq 0$. The buyer learns p and then decides to either buy or not. If the buyer buys, then her utility is $v - p$.

- Strategies for S : $p \in \mathbb{R}_+$.
- Strategies for B : $b : \mathbb{R}_+ \rightarrow \{0, 1\}$, $b(p)$ is a function indicating when the buyer will buy.
- Utility (payoff) for S : $u_S(p, b) = p \times b(p) = \begin{cases} p & b(p) = 1 \\ 0 & b(p) = 0 \end{cases}$
- Utility (payoff) for B : $u_B(p, b) = (v - p) \times b(p) = \begin{cases} v - p & b(p) = 1 \\ 0 & b(p) = 0. \end{cases}$

Example 1.23

Complete Information If the seller knows v , then the only subgame perfect equilibrium is for the seller to choose $p = v$ and for the buyer to buy if and only if $p \leq v$, i.e., $b(p) = I_{\{v \geq p\}}(p)$, where I denotes the indicator function.

Incomplete Information Suppose that the seller doesn't know v , but has beliefs about it that can be represented by a pdf $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$, so that $\Pr(v \in [a, b]) = \int_a^b f(v)dv$. Then the seller's expected utility:

$$u_S(p) = \mathbb{E}(u_S(p, b)) = p \int_0^\infty b(p) f(v) dv = p \int_0^\infty I_{\{v \geq p\}} f(v) dv = p \int_p^\infty f(v) dv = p(1 - F(p))$$

If the seller maximizes $u_S(p)$, we take the FOC:

$$\frac{dU_S}{dp} = 1 - F(p^*) - p^* f(p^*) = 0 \Rightarrow p^* = \frac{1 - F(p^*)}{f(p^*)}.$$

If the above equation has a unique solution p^* , then it is a maximum and p^* is called **the monopoly price**.

Definition 1.24

A **belief space** is a tuple $(N, \Omega, \{T_i\}_{i \in N}, \{\tau_i\}_{i \in N}, \{\mu_i\}_{i \in N})$ where $(N, \Omega, \{T_i\}_{i \in N}, \{\tau_i\}_{i \in N})$ is knowledge space; and μ_i is player i 's belief over Ω . If the μ_i 's are all equal, we say that the players have a **common prior**.

Posterior beliefs. Notice that $\tau_i : \Omega \rightarrow T_i$ become random variables in a belief space. The posterior μ_i , for event A , when conditioned on τ_i , is denoted $\mu_i(A|\tau_i) = \text{Pr}_i(A|\tau_i)$. This expression is a random variable, since τ_i is. When $\tau_i(\omega) = t_i$, we can compute it:

$$\mu_i(A|\tau_i(\omega) = t_i) = \text{Pr}_i(A|\tau_i(\omega) = t_i) = \frac{\mu_i(A \cap \tau_i^{-1}(t_i))}{\mu_i(\tau_i^{-1}(t_i))}.$$

When the underlying knowledge space is given in terms of algebras, we write the above as $\mu_i(A|\Sigma_i) = \text{Pr}_i(A|\Sigma_i)$, and we write:

$$\mu_i(A|\Sigma_i)(\omega) = \text{Pr}_i(A|\Sigma_i)(\omega) = \frac{\mu_i(E \cap P_i(\omega))}{\mu_i(P_i(\omega))}.$$

Example 1.25

Consider two weather forecasters. Both believe that half of the days have rain and the other half do not. They also believe that their forecasts predict the future with some probability $p > 1/2$, and that their forecasts fail or succeed independently of each other. Their common prior μ over Ω is given by the table below.

ω	$\mu(\omega)$	$\tau_1(\omega)$	$\tau_2(\omega)$
RRR	$\frac{1}{2}p^2$	R	R
RRS	$\frac{1}{2}(1-p)^2$	R	R
RSR	$\frac{1}{2}p(1-p)$	R	S
RSS	$\frac{1}{2}p(1-p)$	R	S
SRR	$\frac{1}{2}p(1-p)$	S	R
SRS	$\frac{1}{2}p(1-p)$	S	R
SSR	$\frac{1}{2}(1-p)^2$	S	S
SSS	$\frac{1}{2}p^2$	S	S

- $\Pr(t_2 = R | t_1 = R)$?
- $\Pr(t_2 = R | t_1 = S)$?
- $\Pr(\omega_3 = R | t_1 = R)$?
- $\Pr(\omega_3 = R | t_1 = S)$?

Agreeing to disagree

Let $(N, \omega, \{\mu\}_{i \in N}, \{\Sigma_i\}_{i \in N})$ be a finite belief space with **common priors**, $\mu_i = \mu$ for all i . Let $X : \Omega \rightarrow \mathbb{R}$ be random variable. $E(X)$ denotes the expectation of X with respect to μ . The conditional expectation:

$$E[X|\Sigma_i](\omega) = \frac{\sum_{\omega' \in P_i(\omega)} \mu(\omega') X(\omega')}{\sum_{\omega' \in P_i(\omega)} \mu(\omega')}$$

Theorem 1.26 (Aumann's Agreement Theorem, 1976)

Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Suppose that at ω_0 it is common knowledge that $E[X|\Sigma_i] = q_i$ for $i = 1, \dots, n$ and some numbers $(q_1, \dots, q_n) \in \mathbb{R}^n$. Then $q_1 = \dots = q_n$.

Corollary 1.27

*Aumann proved a version of the theorem where the random variable X was an indicator function of some event A . Thus: if two players have common priors over a finite space, and it is common knowledge that their posteriors for some event A are q_1 and q_2 , then $q_1 = q_2$, i.e., the players **cannot agree to disagree!***

Proof of Theorem 1.26

Recall first the law of total expectation. Let S be an element of an algebra $\Pi \subseteq 2^\Omega$, and let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Then

$$E[X|S] = E[E[X|\Pi]|S].$$

- Milgrom and Stokey used Aumann's Theorem to show that rational agents with common priors can never agree to trade.
- Setup:
 - Two agents, one has an indivisible good that she might be interested to sell to the second.
 - The good can be sold tomorrow at an auction for an unknown price between \$0 and \$ 1,000, in integer increments.
 - Let $\Omega = T_1 \times T_2 \times T$ and let μ be the common prior. $T = \{\$0, \$1, \dots, \$1000\}$ represents the auction price, and $T_1 \times T_2$ is some finite set that describes possible events that may influence the good of the price.
 - Players' types are given by $\tau_i(\omega_1, \omega_2, \omega_3) = \omega_i$ for $i = 1, 2$.
 - $P : \Omega \rightarrow T$ is the auction price with $P(\omega) = \omega_3$.
 - Q_1 is player 1's conditional expected price: $Q_1 = E[P|\tau_1]$.
 - Q_2 is player 2's conditional expected price: $Q_2 = E[P|\tau_2]$.

Example 1.28

A specific instance of the setup in the previous slide: let $T_1 = T_2 = T$ and let μ be the uniform distribution over $\{(\omega_1, \omega_2, \omega_3) : \omega_3 = \omega_1 \text{ or } \omega_3 = \omega_2\}$.

The No-Trade Theorem

Theorem 1.29 (Milgrom and Stokey, 1982)

If at some $\omega \in \Omega$ it is common knowledge that $Q_1 \leq q \leq Q_2$, then $Q_1(\omega) = q = Q_2(\omega)$.

Geanakoplos and Plemarchakis show that repeatedly communicating posteriors leads agents to converge to a common posterior, which is then common knowledge.

Theorem 1.30 (Geanakoplos and Polemarchakis, 1982)

Let (Ω, Σ, μ) be a probability space. Fix $A \in \Sigma$. Let X_1 and X_2 be two random variables on this space. Denote $Q_1^0 = \mu(A|X_1)$ and $Q_2^0 = \mu(A|X_2)$. For $t = 1, 2, \dots$, let

$$Q_1^t = \mu(A|X_1, Q_2^0, Q_2^1, \dots, Q_2^{t-1}) \quad Q_2^t = \mu(A|X_2, Q_1^0, Q_1^1, \dots, Q_1^{t-1}).$$

Then $\lim_t Q_1^t$ and $\lim_t Q_2^t$ (almost-surely) exists and are equal.